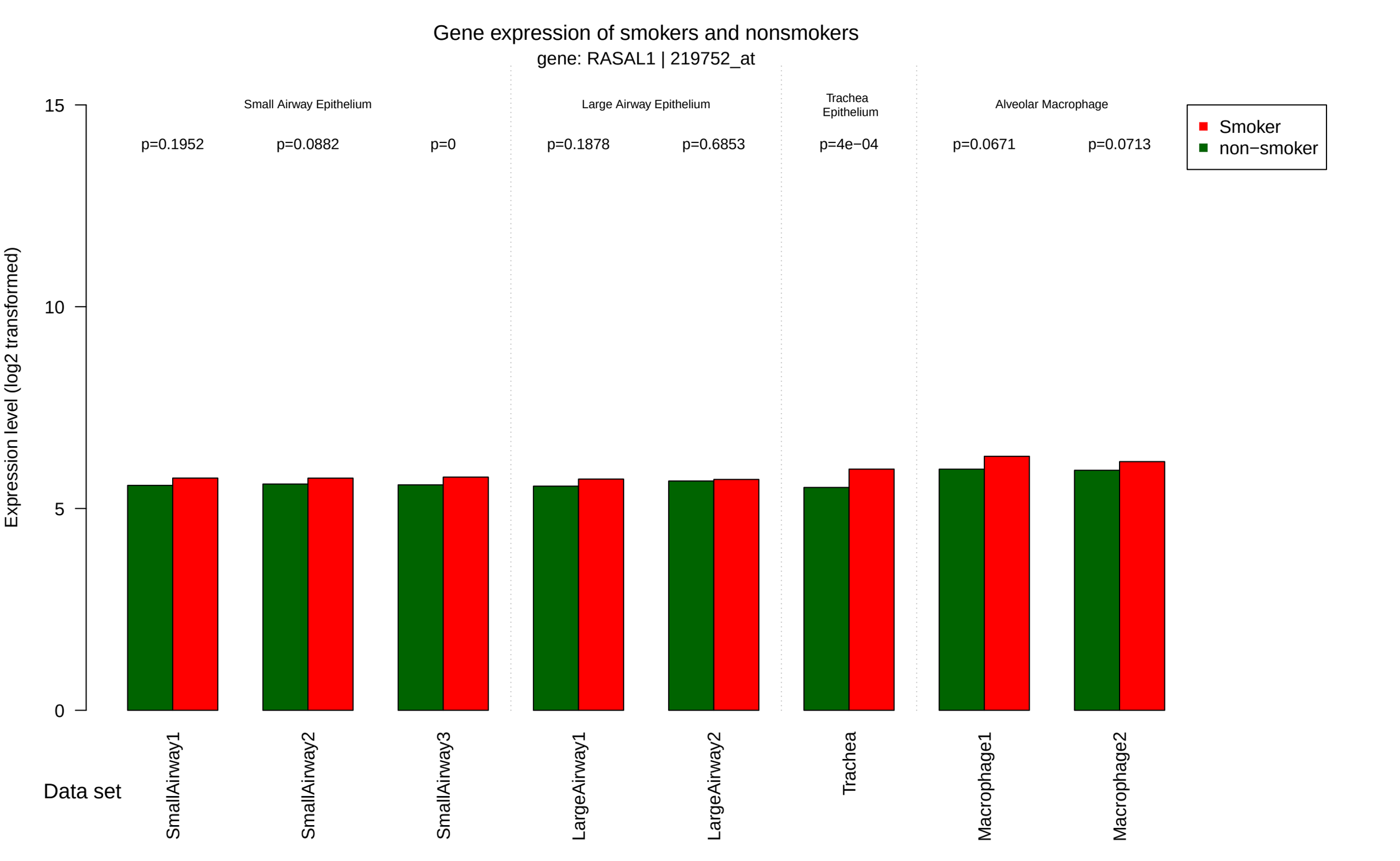
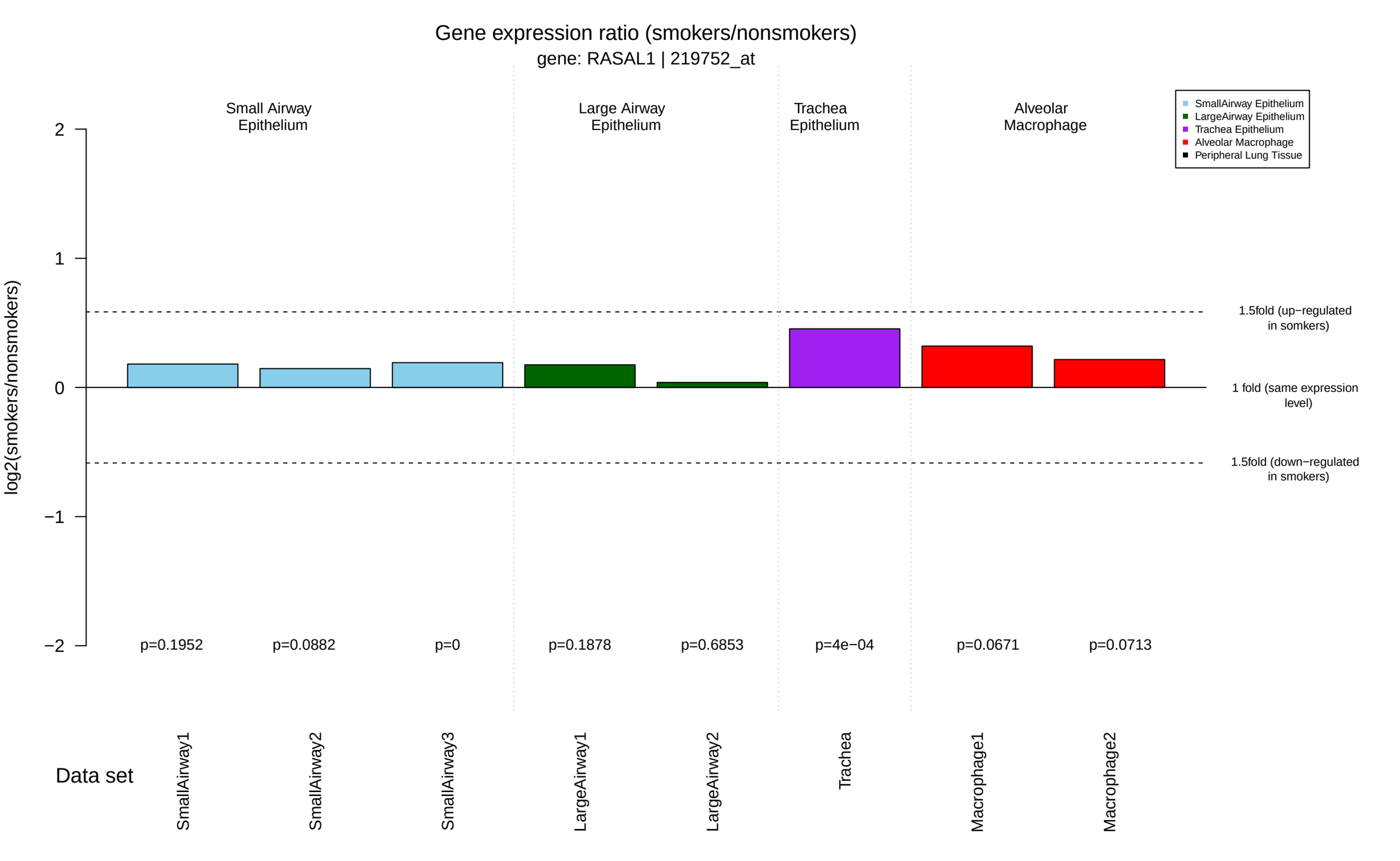
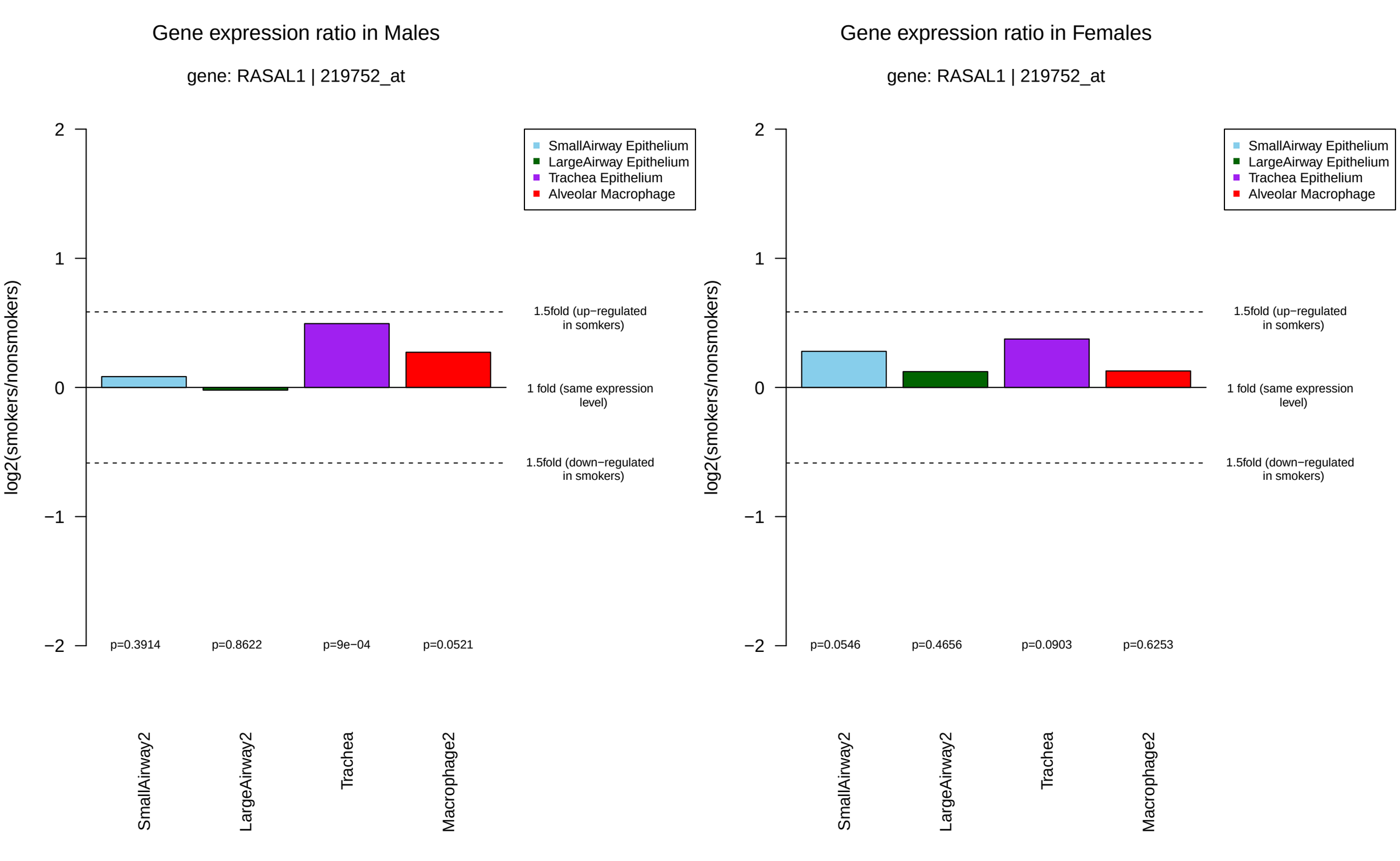
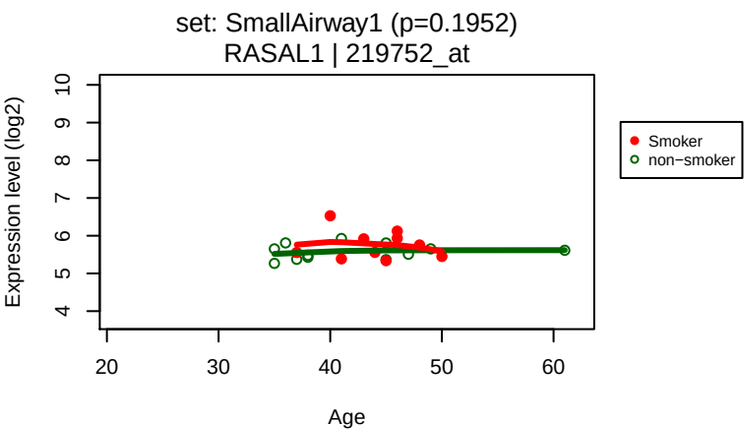
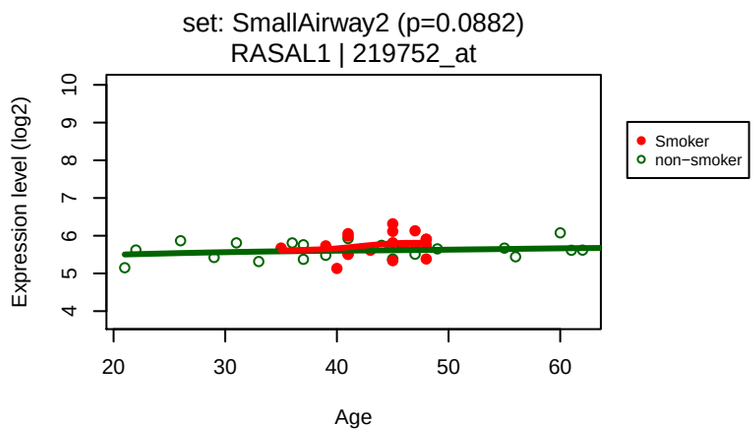
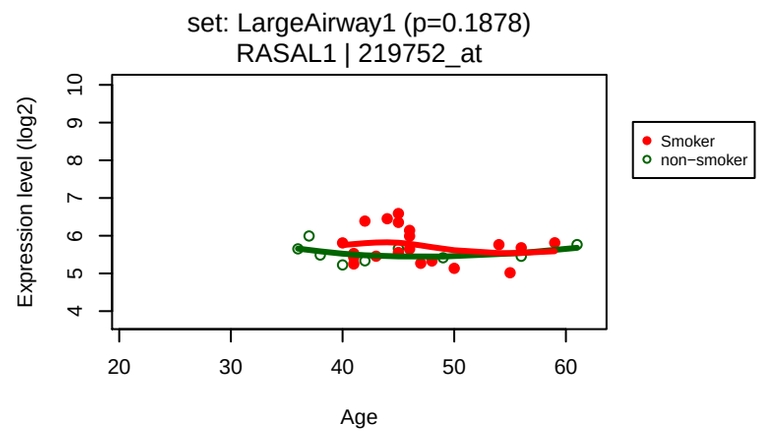

Gene expression of smokers and nonsmokers
gene: RASAL1 | 219752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1952
p=0.0882
p=0
p=0.1878
p=0.6853
p=4e−04
p=0.0671
p=0.0713
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

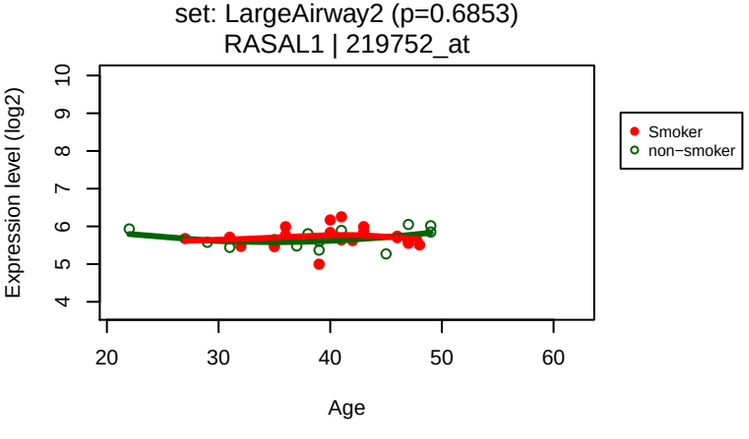

Gene expression ratio (smokers/nonsmokers)
gene: RASAL1 | 219752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1952
p=0.0882
p=0
p=0.1878
p=0.6853
p=4e−04
p=0.0671
p=0.0713
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

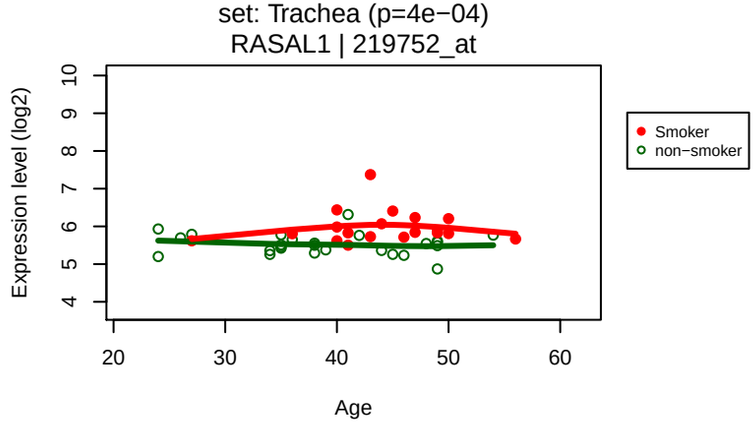

Gene expression ratio in Males
gene: RASAL1 | 219752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3914
p=0.8622
p=9e−04
p=0.0521
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL1 | 219752_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0546
p=0.4656
p=0.0903
p=0.6253
SmallAirway2
LargeAirway2
Trachea
Macrophage2

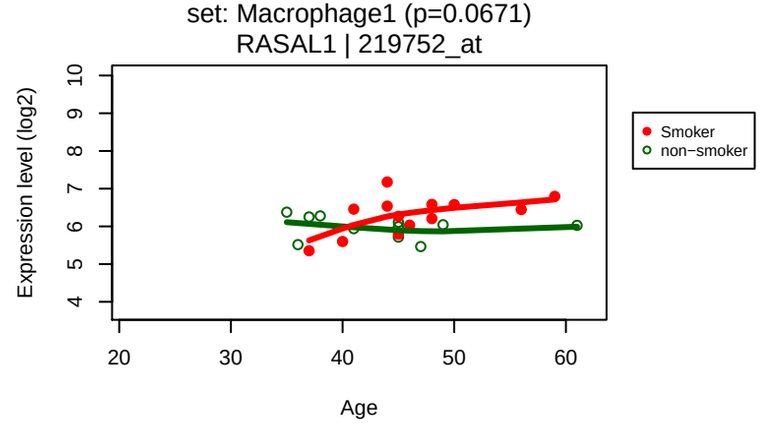

set: SmallAirway1 (p=0.1952)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker

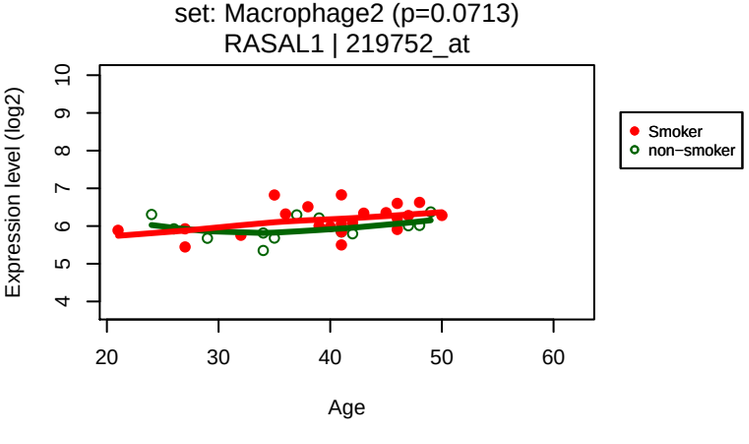

set: SmallAirway2 (p=0.0882)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker

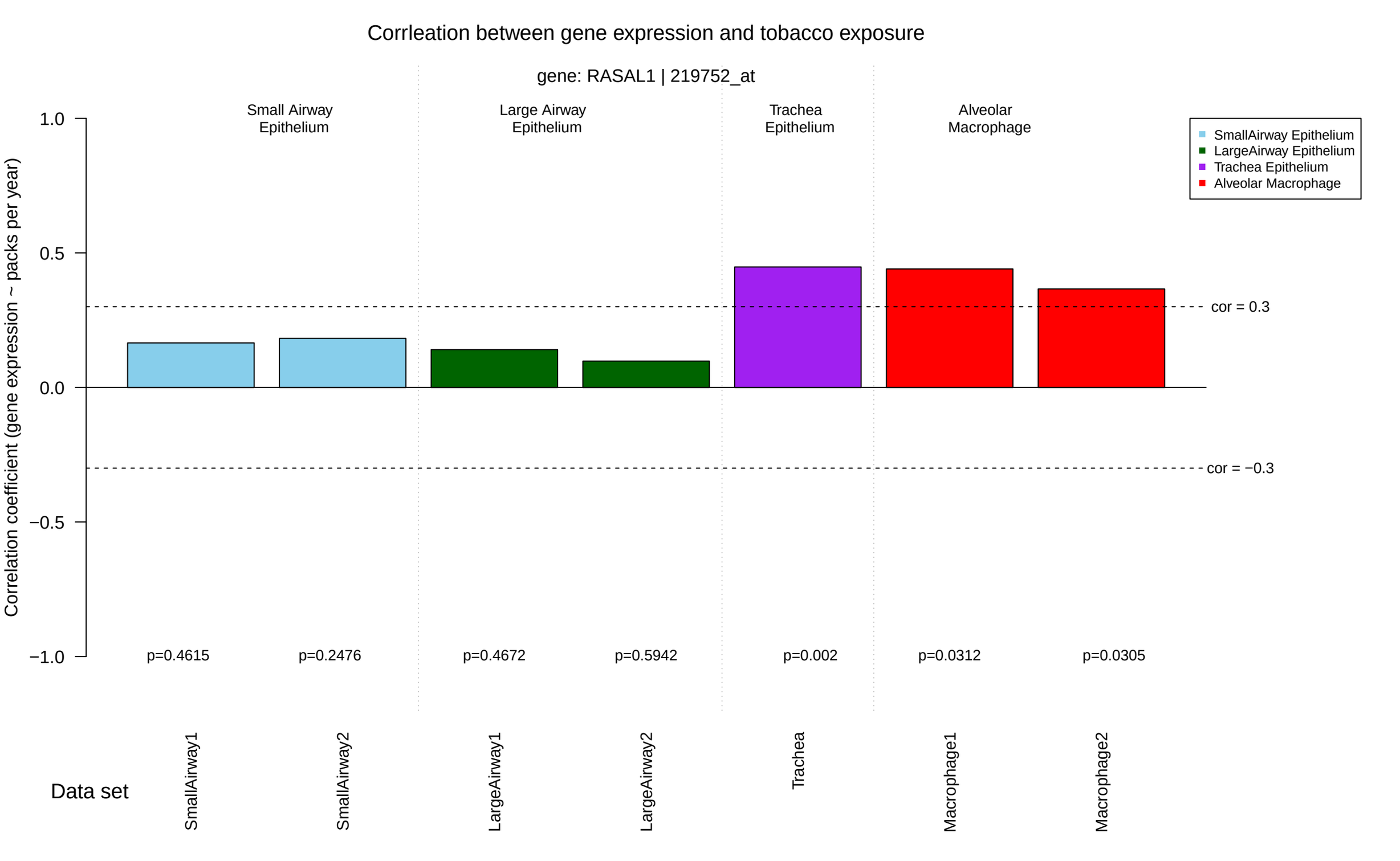

set: LargeAirway1 (p=0.1878)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6853)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0671)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0713)
RASAL1 | 219752_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL1 | 219752_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4615
p=0.2476
p=0.4672
p=0.5942
p=0.002
p=0.0312
p=0.0305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2